

# Integration by Substitution

## Today's Goals

- ◆ You should be able to explain the process of integration by substitution in relation to the Chain Rule.
- ◆ You should be able to use substitution to evaluate definite and indefinite integrals.
- ◆ You should be able to determine whether a particular substitution is helpful in evaluating an integral.

1. Can you evaluate the following integrals?

(a)  $\int_1^4 \frac{1}{3} \sqrt{u} \, du$

### Solution

An antiderivative of  $\sqrt{u} = u^{1/2}$  is  $\frac{2}{3}u^{3/2}$ , so

$$\begin{aligned} \int_1^4 \frac{1}{3} \sqrt{u} \, du &= \frac{1}{3} \cdot \frac{2}{3} u^{3/2} \Big|_1^4 \\ &= \frac{2}{9} \cdot 4^{3/2} - \frac{2}{9} \cdot 1^{3/2} \\ &= \frac{2}{9} \cdot 8 - \frac{2}{9} \cdot 1 \\ &= \boxed{\frac{14}{9}} \end{aligned}$$

(b)  $\int \cos(3x) \, dx$

### Solution

You might guess that  $\sin(3x)$  is involved in an antiderivative of  $\cos(3x)$ . The derivative of  $\sin(3x)$  is  $3\cos(3x)$ , so the derivative of  $\frac{1}{3}\sin(3x)$  is  $\cos(3x)$ . Therefore,  $\int \cos(3x) \, dx =$

$$\boxed{\frac{1}{3} \sin(3x) + C}.$$

(c)  $\int \cos(x^2) \, dx$

### Solution

There is no way to write down this indefinite integral! It turns out that  $\cos(x^2)$  is an example of a function whose antiderivatives cannot be expressed as a finite sum of familiar functions (like exponentials, logs, trig functions, etc.). This doesn't mean that  $\cos(x^2)$  doesn't have antiderivatives (FTOC 1 guarantees that it does!), but we can't write down any sort of nice formula for those antiderivatives.

2. Fill in the blank in each statement:

$$(a) \int \frac{\cos(u)}{\sin(u)} du = \ln|\sin(u)| + C$$

#### Solution

The derivative of  $\sin(u)$  is  $\cos u$ .

$$(b) \int \frac{\cos(x^2) \cdot 2x}{\sin(x^2)} dx = \ln|\sin(x^2)| + C$$

#### Solution

By the Chain Rule, the derivative of  $\sin(x^2)$  is  $\cos(x^2) \cdot 2x$ .

### Integration by Substitution

If  $f$  is a continuous function with antiderivative  $F$  and  $g$  is a differentiable function, then

$$\int f(g(x)) \cdot g'(x) dx = F(g(x)) + C$$

We often write  $u = g(x)$  and  $du = g'(x) dx$ , in which case the formula is

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du$$

3. Evaluate the following integrals using substitution.

$$(a) \int \frac{\sin(\ln x)}{5x} dx$$

#### Solution

It looks like a good substitution is  $u = \ln x$ , since it's the inner function in the composition  $\sin(\ln x)$ , and we also see  $du = \frac{1}{x} dx$  in the integral. So,

$$\begin{aligned} \int \frac{\sin(\ln x)}{5x} dx &= \int \frac{\sin u}{5} du \\ &= -\frac{1}{5} \cos u + C \\ &= -\frac{1}{5} \cos(\ln x) + C \end{aligned}$$

$$(b) \int e^{-2x} \cos(e^{-2x}) dx$$

#### Solution

It looks like a good substitution is  $u = e^{-2x}$  because it's the inner function in the composition

$\cos(e^{-2x})$ , and we also see something related to  $du = -2e^{-2x} dx$ , namely  $-\frac{du}{2} = e^{-2x} dx$ .

$$\begin{aligned}\int e^{-2x} \cos(e^{-2x}) dx &= \int \cos u \left( -\frac{du}{2} \right) \\ &= -\frac{1}{2} \int \cos u du \\ &= -\frac{1}{2} \sin u + C \\ &= \boxed{-\frac{1}{2} \sin(e^{-2x}) + C}\end{aligned}$$

(c)  $\int_1^4 x^2 \sqrt{1+x^3} dx$

### Solution

It looks like a good substitution is  $u = 1 + x^3$  since it's the inner function in the composition  $\sqrt{1+x^3}$ , and we also see something related to  $du = 3x^2 dx$ , namely  $\frac{1}{3} du = x^2 dx$ . When  $x = 1$ ,  $u = 1 + 1^3 = 2$ ; when  $x = 4$ ,  $u = 1 + 4^3 = 65$ . So,

$$\begin{aligned}\int_1^4 x^2 \sqrt{1+x^3} dx &= \int_2^{65} \sqrt{u} \cdot \frac{1}{3} du \\ &= \frac{1}{3} \int_2^{65} u^{1/2} du \\ &= \frac{2}{9} u^{3/2} \Big|_2^{65} \\ &= \boxed{\frac{2}{9} (65^{3/2} - 2^{3/2})}\end{aligned}$$

*Note on notation:* If you prefer not to change the bounds of the integral to be in terms of  $u$ , your first line should instead read

$$\int_1^4 x^2 \sqrt{1+x^3} dx = \int_{x=1}^{x=4} \sqrt{u} \cdot \frac{1}{3} du.$$

It is **not** correct to write

$$\int_1^4 x^2 \sqrt{1+x^3} dx = \int_1^4 \sqrt{u} \cdot \frac{1}{3} du.$$

After all, we saw in #1(a) that  $\int_1^4 \sqrt{u} \cdot \frac{1}{3} du$  is equal to  $\frac{14}{9}$ , and we've just seen that  $\int_1^4 x^2 \sqrt{1+x^3} dx \neq \frac{14}{9}$ .

(d)  $\int \sin x \cos^5 x dx$

### Solution

It looks like a good substitution is  $u = \cos x$  because it's the inner function in the composition

$\cos^5 x$ , and we see something related to  $du = -\sin x \, dx$  in the integral, namely  $-du = \sin x \, dx$ .

$$\begin{aligned}\int \sin x \cos^5 x \, dx &= \int (\sin x)(\cos x)^5 \, dx \\ &= \int u^5 (-du) \\ &= -\int u^5 \, du \\ &= -\frac{1}{6}u^6 + C \\ &= \boxed{-\frac{1}{6}\cos^6 x + C}\end{aligned}$$

(e)  $\int_{1/e}^{e^3} \frac{4 \ln x}{x} \, dx$

### Solution

It looks like  $u = \ln x$  is a promising substitution because we also see  $du = \frac{1}{x} \, dx$  in the integral:

$$\begin{aligned}\int_{1/e}^{e^3} \frac{4 \ln x}{x} \, dx &= \int_{-1}^3 4u \, du \\ &= 2u^2 \Big|_{-1}^3 \\ &= 2 \cdot 3^2 - 2 \cdot (-1)^2 \\ &= \boxed{16}\end{aligned}$$

(f)  $\int \frac{e^x}{2e^x + 5} \, dx$

### Solution

It looks like  $u = 2e^x + 5$  is a good substitution since it's the denominator and we also see something related to  $du = 2e^x \, dx$  in the integral, namely  $\frac{du}{2} = e^x \, dx$ :

$$\begin{aligned}\int \frac{e^x}{2e^x + 5} \, dx &= \int \frac{1}{u} \cdot \frac{du}{2} \\ &= \int \frac{1}{2} \cdot \frac{1}{u} \, du \\ &= \frac{1}{2} \ln |u| + C \\ &= \boxed{\frac{1}{2} \ln |2e^x + 5| + C}\end{aligned}$$

(g)  $\int_0^1 \sqrt{5x+4} \, dx$

### Solution

It looks like  $u = 5x + 4$  is a good substitution since it's the inner function in the composition  $\sqrt{5x + 4}$ . Then,  $du = 5 dx$ , so  $\frac{du}{5} = dx$ . As  $x$  goes from 0 to 1,  $u = 5x + 4$  goes from  $u = 5 \cdot 0 + 4 = 4$  to  $5 \cdot 1 + 4 = 9$ :

$$\begin{aligned} \int_0^1 \sqrt{5x+4} dx &= \int_4^9 \sqrt{u} \frac{du}{5} \\ &= \frac{1}{5} \int_4^9 u^{1/2} du \\ &= \frac{1}{5} \cdot \frac{2}{3} u^{3/2} \Big|_4^9 \\ &= \frac{2}{15} (9^{3/2} - 4^{3/2}) \\ &= \frac{2}{15} (27 - 8) \\ &= \boxed{\frac{38}{15}} \end{aligned}$$

4. One of the following integrals can easily be evaluated using substitution. Which one? Evaluate it.

(a)  $\int \frac{1}{e^{x^2}} dx$

(b)  $\int \frac{x}{e^{x^2}} dx$

### Solution

Both integrals involve a composition with  $u = x^2$  as the inner function, but only the second integral has something related to  $du = 2x dx$ , namely  $\frac{du}{2} = x dx$ . That's the one we can easily evaluate with substitution:

$$\begin{aligned} \int \frac{x}{e^{x^2}} dx &= \int \frac{1}{e^u} \left( \frac{du}{2} \right) \\ &= \frac{1}{2} \int e^{-u} du \\ &= -\frac{1}{2} e^{-u} + C \\ &= \boxed{-\frac{1}{2} e^{-x^2} + C} \end{aligned}$$

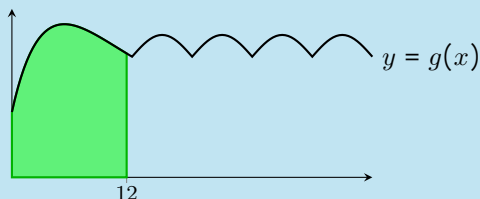
The other indefinite integral,  $\int \frac{1}{e^{x^2}} dx$ , cannot be easily evaluated using substitution. In fact, like in #1(c), the antiderivatives of  $\frac{1}{e^{x^2}}$  cannot be expressed as a finite sum of familiar functions (like exponentials, logs, trig functions, etc.), so  $\int \frac{1}{e^{x^2}} dx$  cannot be easily evaluated with any method!

5. Suppose  $g$  is a mysterious function, and the only thing you know about it is that  $\int_0^{12} g(x) dx = \pi$ . For each of the following integrals, decide whether you have enough information to evaluate the integral; if so, evaluate it.

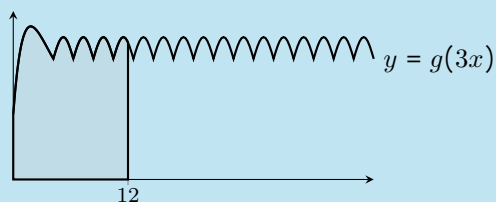
(a)  $\int_0^{12} g(3x) dx$

## Solution

There is not enough information. (As you can see in the next part, substituting  $u = 3x$  doesn't help.) We can understand why graphically; suppose  $g$  looks like this. The only information we're given is that the green area is  $\pi$ :



The graph of  $g(3x)$  is stretched horizontally to be  $\frac{1}{3}$  as wide, so here's how we picture  $\int_0^{12} g(3x) dx$ :



But there's no way for us to find this area because we can't express it simply in terms of the green area above, which is the only area we know.

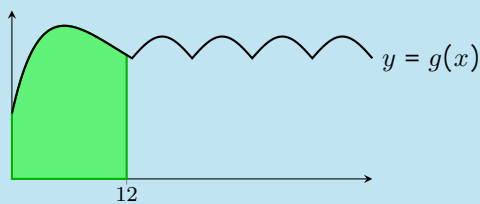
(b)  $\int_0^4 g(3x) dx$

## Solution

We can evaluate this by substituting  $u = 3x$ ; then,  $du = 3 dx$ , so  $\frac{du}{3} = dx$ . Since  $x$  goes from 0 to 4,  $u = 3x$  goes from 0 to 12; thus,

$$\begin{aligned} \int_0^4 g(3x) dx &= \int_0^{12} g(u) \frac{du}{3} \\ &= \frac{1}{3} \int_0^{12} g(u) du \\ &= \boxed{\frac{\pi}{3}} \end{aligned}$$

This should also make sense graphically; the graph of  $g(3x)$  is obtained by stretching the graph of  $g(x)$  horizontally by a factor of  $\frac{1}{3}$ :





Therefore, the area under  $g(3x)$  from  $x = 0$  to  $x = 4$  should be  $\frac{1}{3}$  as large as the area under  $g(x)$  from  $x = 0$  to  $x = 12$ .

(c)  $\int_0^{144} g(x^2) dx$

#### Solution

There is not enough information. (If you try substituting  $u = x^2$ , you instead end up with the integral in (d).)

(d)  $\int_0^{2\sqrt{3}} xg(x^2) dx$

#### Solution

We can evaluate this using the substitution  $u = x^2$ ; then,  $du = 2x dx$ , so  $\frac{du}{2} = x dx$ . As  $x$  goes from 0 to  $2\sqrt{3}$ ,  $u = x^2$  goes from  $0^2 = 0$  to  $(2\sqrt{3})^2 = 12$ :

$$\begin{aligned} \int_0^{2\sqrt{3}} xg(x^2) dx &= \int_0^{12} g(u) \frac{du}{2} \\ &= \frac{1}{2} \int_0^{12} g(u) du \\ &= \boxed{\frac{\pi}{2}} \end{aligned}$$

(e)  $\int_0^{144} xg(x^2) dx$

#### Solution

There is not enough information.